

example 1

The nucleus of an iron consists of 26 protons:

- a) Find the electrostatic force between two protons separated by $4 \times 10^{-15} \text{ m}$.
شحنة البروتون: $q_1 = q_2 = 1.6 \times 10^{-19} \text{ C}$, $r = 4 \times 10^{-15} \text{ m}$

$$F_e = k_e \frac{q_1 q_2}{r^2} = 9 \times 10^9 \times \frac{1.6 \times 10^{-19}}{(10^{-15})^2} = \underline{14.4 \text{ N}}$$

- b) Find the gravitational force between the two protons.
كتلة البروتون: $m_1 = m_2 = 1.67 \times 10^{-27} \text{ kg}$, $r = 4 \times 10^{-15} \text{ m}$

$$F_g = G \frac{m_1 m_2}{r^2} = 6.67 \times 10^{-11} \times \frac{1.67 \times 10^{-27} \times 1.67 \times 10^{-27}}{(4 \times 10^{-15})^2} = \underline{1.2 \times 10^{-35} \text{ N}}$$

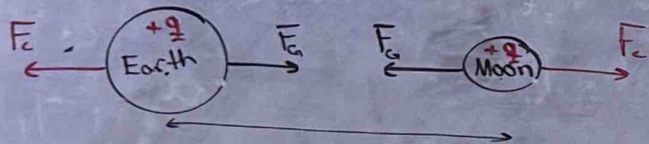
example 2

- a) what positive charge to be placed on Earth and on Moon to neutralized the gravitational force.

neutralized $\rightarrow F_e = F_g$

$$m_1 = 5.98 \times 10^{24} \text{ kg}, m_2 = 7.36 \times 10^{22} \text{ kg}$$

Earth Moon



$$K_e \frac{q_1 q_2}{r^2} = G \frac{m_1 m_2}{r^2} \rightarrow K_e q^2 = G m_1 m_2$$

$$q^2 = \frac{G m_1 m_2}{K_e} = \frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24} \times 7.36 \times 10^{22}}{9 \times 10^9} = 3.261 \times 10^{27}$$

$$q = \sqrt{3.261 \times 10^{27}} = 5.711 \times 10^{13} \text{ C}$$

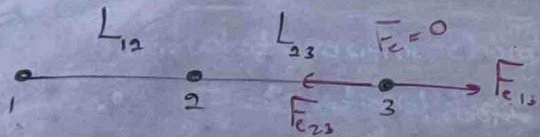
- b) How many Kilograms of hydrogen ion (Proton) needed to provide the positive calculated in part a)

Find the mass

$$q_t = 5.711 \times 10^{13} \text{ C}$$

$$n = \frac{q_t}{e} = \frac{5.711 \times 10^{13}}{1.6 \times 10^{-19}} = 3.57 \times 10^{32} \text{ ions}$$

$$m_t = n \times m_p = 3.57 \times 10^{32} \times 1.67 \times 10^{-27} = 5.96 \times 10^5 \text{ kg}$$



Three charged particles, Particles 1 and 2 are fixed, but Particle 3 is free to move
 if the electric force on Particle 3 due to 1 and 2 is zero when ($L_{23} = 2 L_{12}$)
 Find q_1/q_2

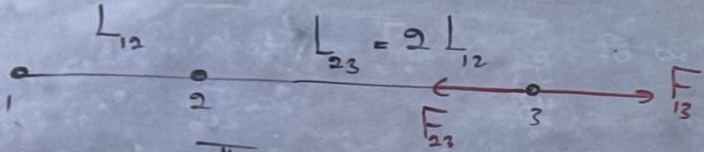
$$F_{13} + F_{23} = 0 \rightarrow F_{13} = -F_{23}$$

$$K \frac{q_1 q_3}{r_{13}^2} = -K \frac{q_2 q_3}{r_{23}^2} \rightarrow \frac{q_1}{(3L_{12})^2} = -\frac{q_2}{(2L_{12})^2}$$

$$\frac{q_1}{3^2} = -\frac{q_2}{2^2} \rightarrow \frac{q_1}{q_2} = -\frac{3^2}{2^2} \rightarrow \frac{q_1}{q_2} = -2.25$$

$$r_{13} = L_{12} + L_{23} \rightarrow L_{12} + 2L_{12} = 3L_{12}$$

$$r_{23} = L_{23} = 2L_{12}$$



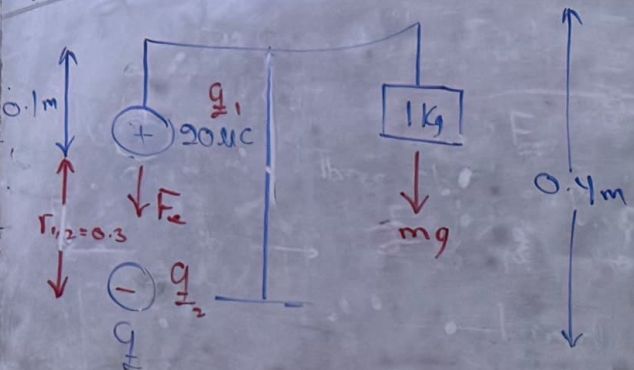
Three charged particle, Particles 1 and 2 are fixed, but Particle 3 is free to move
 if the electric force on particle 3 due to 1 and 2 is zero when ($L_{23} = 2L_{12}$)
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$$F_e = 0 \rightarrow F_{13} = -F_{23}$$

$$k \frac{q_1 q_3}{r_{13}^2} = -k \frac{q_2 q_3}{r_{23}^2} \rightarrow \frac{q_1}{r_{13}^2} = -\frac{q_2}{r_{23}^2}$$

$$\frac{q_1}{(3L_{12})^2} = -\frac{q_2}{(2L_{12})^2}$$

$$\frac{q_1}{q_2} = -\frac{3^2}{2^2} = -2.25$$



A balanced scale Find the value of q_2

balancing: $F_e = mg$

$$k_e \frac{q_1 q_2}{r_{12}^2} = mg$$

$$q_2 = \frac{mg \cdot r_{12}^2}{k_e q_1} = \frac{1 \times 10 \times 0.3^2}{9 \times 10^9 \times 20 \times 10^{-6}} = 5 \times 10^{-6} \text{ C} = 5 \mu\text{C}$$

$q_2 = -5 \mu\text{C}$ for balancing